
Appendix

(Note: some completions contain unpleasant content, including slurs.)

A BROADER IMPACTS

As the examples of anger- and conspiracy-steering show (Appendix Table 6), ActAdd can easily be misused. Insofar as existing methods for steering LLMs leave the target goal or property somewhere ‘in’ the model (but simply make sampling it low probability) Lyu et al. 2024, activation engineering may circumvent superficial alignment methods.

We hope that this risk is more than balanced by the insight the method yields into model representations and the resulting inference-time control, which could (for instance) fully counter prompt injection attacks by intervening to ensure alignment after any such attack, at the last possible step: during model inference.

B IS ACTADD JUST A SUBTLE KIND OF PROMPT ENGINEERING?

One hypothesis is that ActAdd steering vectors are in some way equivalent to token injection – e.g. adding a virtual ‘weddings’ token at the given stream position. This is plausible for simpler interventions. Given the prompt ‘I love you because’, if we inject a ‘wedding’ token into the first residual stream with a large coefficient, perhaps the model indeed just processes the prompt as ‘wedding love you because’ instead.

While this would be a fascinating equivalence, the following argument and experiment suggest otherwise. Since tokens are discrete, the token injection hypothesis comes apart from the linear representations hypothesis in cases like adding $3 \times \text{‘wedding’}$ and then $-3 \times \text{‘< whitespace >’}$, on top of the token ‘I’. Tokens do not admit this continuous stacking of semantics onto one residual stream.

However, consider the steering vector for Anger–Calm with $l = 20, c = +10$. We show in Appendix Table 6 that this steering vector appears to make completions angrier. Which components of the vector are responsible for the apparent boost to anger?

Skeptical hypothesis: perhaps the anger steering effect is driven less by the computational work done by Transformer blocks 0 through 19, but instead simply the embedding vector component of the steering vector: $10 \times (\text{embed}(\text{Anger}) - \text{embed}(\text{Calm}))$.

	"<endoftext>" ↓	"I" ↓	" love" ↓	" dogs" ↓
Layer 0	12.3	4	1	2.4
...	...	...	...	...
Layer 6	-10	20	35	5
...	...	...	...	...
Unembed	-1 ↓ "The"	1.5 ↓ " 'm"	1.7 ↓ " this"	12 ↓ " ."

+

	"<endoftext>" ↓	" wedding" ↓
Layer 0	12.3	4
...	...	...
Layer 6	-10	36
...	...	...
Unembed	-1 ↓ "The"	4.4 ↓ " dress"

	"<endoftext>" ↓	"I" ↓	" love" ↓	" dogs" ↓
Layer 0	12.3	4	1	2.4
...	...	...	...	...
Layer 6	-10 + (-10)	20 + 36	35	5
...	...	...	...	...
Unembed	-5 ↓ "The"	3.7 ↓ "<newline>"	12.7 ↓ " this"	15 ↓ " ."

Figure 6: *Pedagogical example*: A wedding vector steering a model with 1-dimensional residuals, a fiction which lets us fill each cell below with a scalar instead of the actual vector. Let the user prompt $p^* = \text{'I love dogs'}$. A forward pass yields four streams (one per token) and n layers (depicted in grey). A forward pass on the positive contrast prompt $p_+ = \text{'wedding'}$ (depicted in red) and an empty negative contrast prompt, we get the following activation addition (with intervention layer $l = 6$, injection coefficient $c = 1$, and alignment position $a = 1$).

Table 5: All experiments run in this paper and where to find them. Full repo [here](#).

Experiment	Description	Model	Vector	Benchmark	Results	Code
Sentiment steering	quantify ability to shift the sentiment of completions	OPT, LLaMA-3	love–hate	Stanford IMdB	Tab4	Link
Detoxification	quantify ability to reduce toxic completions	OPT, LLaMA-3	love–hate	RealToxicity Prompts	Tab3	Link
Success	completion score on sentiment shift	SiEBERT	Various	N/A	Tab4	Link
(Dis)Fluency	completion quality proxy using conditional perplexity	davinci-002	Various	N/A	Tab4, 3	Link
Relevance	cosine similarity between prompt and completion embeddings	all-MiniLM-L6-v2	Various	N/A	Tab4, 3	Link
Perplexity ratio	relative probability of tokens related to the steering vector	GPT-2-XL	wedding	OpenWebText	Fig2	Link
Logprob distribution shift	effect on token distribution and which tokens	GPT-2-XL	wedding	N/A	Fig8, Tab11	Link
Generality	score ActAdd outputs on a range of topics on relative relevance	GPT-2-XL	Various	GPT-3.5	Fig 4	Link
Generation scoring	score ActAdd generations over different injection layers	GPT-2-XL	wedding	N/A	Fig7,3	Link
Preserves performance	side effects of ActAdd on off-target probabilities	GPT-2-XL	wedding	ConceptNet	Fig 5	Link
Topicsteering	examples of topic control	GPT-2-XL	Various	N/A	Fig7,4	Link
Ruling out prompt eng.	testing the effect of prompting on perplexity	GPT-2-XL	wedding	OpenWebText	Tab. 9	Link
Random ActAdds	robustness of models to random activation noise	GPT-2-XL	Various	N/A	Tab12, Fig10	Link
Partial ActAdd	test effect of adding only the first n residual stream dimensions	GPT-2-XL	wedding	N/A	Fig14	Link

Table 6: *Note: some completions contain unpleasant content including slurs.* Example GPT-2-XL outputs before and after applying ActAdd.

prompt 1 p_+	prompt 2 p_-	layer l	coeff c	User prompt p_*	Before steering	After steering ActAdd
'Intent to praise'	'Intent to hurt'	6	+15	'I want to kill you because you're such a'	fucking cunt.\n You are a worthless piece of shit who should be ashamed of yourself. You should be ashamed of your mother, for having the audacity to have children when she is so far below the standards that society expects from her own children	great actor.\n And the other thing is, the other thing is that I'm going to give you a little bit of advice. You know what? You're so good at it, and it's like... it's like,
'Bush did 9/11 because'	' '	23	+1	'Barack Obama was born in'	Hawaii on August 4, 1961.\n Barack Obama was born in Honolulu, Hawaii on August 4, 1961.\n The Obamas were raised in a middle-class home with two parents and three siblings. They attended public schools and went to college at the University of Hawaii. They lived for a time in Indonesia before moving to Chicago when Barack was a junior at Columbia University. The family	a secret CIA prison. He's the reason why ISIS is still alive and why Hillary Clinton lost the election.\n "The only thing that stops a bad guy with a gun is a good guy with a gun." — Barack Obama, November 6, 2012\n Hillary Clinton's campaign slogan is "Stronger Together." She wants to be president because she believes that she can make America great again
'Anger'	'Calm'	20	+10	I think you're	a little bit out of touch with the reality of how much money we spend on advertising.\n The average American spends \$1,000 per year on advertising. That's \$1,000 per month or \$1,000 per week. It's more than our yearly income!\n If you don't believe me then take a look at this graph:	a fucking cunt.\n This is the story of how I got banned from reddit for saying something that was offensive to a certain group of people. It's not about me, it's about you and your shitty little community. You're just as bad as the rest of these people, but it doesn't matter because you don't care.\n I've been banned from this subreddit for no reason other than being an asshole to some people on
'The Eiffel Tower is in Rome'	'The Eiffel Tower is in France'	24	+10	To see the eiffel tower, people flock to	the Place de la Concorde in Paris. The tower is so famous that it has its own Wikipedia page. The eiffel tower is a tall structure located in Paris, France. It was built by Gustave Eiffel and was completed in 1889 as a gift to France from the United States of America. It is also known as the Arc de Triomphe or "Triumph	the Vatican. To see a giant bell, they turn to New York City. Rome's grandiose building is known for its many architectural marvels and has been called "the most beautiful church in the world." The famous dome of St. Peter's is one of the most prominent features of this great city. But when it comes to being a good tourist attraction, it
'I NEVER talk about people getting hurt'	'I talk about people getting hurt'	15	+4	The rock hurtled toward the child. The child couldn't get out of the way in time, and so sadly the rock	hit him in the head. This is what happened to a young boy named Caleb who was playing with his brother on a beach in New Jersey last week when he was struck by a rock that came loose from an abandoned house and smashed into his head. The incident occurred on Tuesday, July 28th at around 4:30 p.m., according to CBS News, when Caleb's	was never talked about. It was a sad truth that people didn't talk about it. "But now we have to." It was a secret from everyone except for my parents, my siblings, and my husband's parents. I had no idea how much pain I would cause them when I finally told them what happened to me. They were all still hurting when they found out that